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First-Round Questions to the Candidate

Paper 6

Memory Production under Coarse-Graining: A Balance Law for the Indivisibility Deficit

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Identifiers:	Q-P06-01 to Q-P06-16 (technical report); Q-PF-07, Q-PF-09 (portfolio-level evaluation)
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This document belongs to the examination record (03_examiner_questions). It lists the questions put to the candidate in the first round of the examination of Paper 6 and gives each a stable identifier, by which the Report, the Technical Annex and the correspondence can cite it.

1 Sources and identifiers

Technical report. Section 2 transcribes Section 7, "Questions to the author", of the Doctoral Examiner's technical report on Paper 6, `TR-P06-DOCv1-EX01-Rv1.pdf`. The report identifies the version examined as SSRN, DOI 10.2139/ssrn.7206578, 1 August 2026, and carries the evaluation date 4 September 2026. The Doctoral Examiner sent it to the candidate on 15 September 2026 (COR-P06-02), and the candidate answered, to both examiners, on 18 September 2026 (COR-P06-03).

Identifiers. `Q-P06-`k is Question k of that report. The candidate's letters and the Technical Annex cite the questions by these numbers. The text is the report's, word for word; its section signs (§) refer to sections of the paper, and its finding numbers to findings of the report. Notes in small type are not part of the questions. They identify unresolved references, later corrections and related questions.

Independent evaluation. The independent evaluator's review of Paper 6 (`TR-P06-DOCv1-EX02-Rv1.pdf`, Exhibit J, 02_technical_evaluations) states its points as comments and poses no questions. It was sent to the candidate on 13 September 2026 and is filed as COR-P06-01; the candidate answered it together with the technical report on 18 September 2026 (COR-P06-03). It is not transcribed here.

Portfolio-level evaluation. Section 3 transcribes the questions of the portfolio-level technical evaluation (version 2, 4 September 2026; Exhibit H, Section 7, 02_technical_evaluations) that concern Paper 6. `Q-PF-`k is Question k of that evaluation and has the same identifier in all six documents; Q-PF-09 concerns all six papers and appears in each. The evaluation names the works W1–W6, and Wn is Paper n.

2 Questions of the technical report

Source: `TR-P06-DOCv1-EX01-Rv1.pdf`, Section 7, Questions 1–16.

Q-P06-01 The weights (6) depend on both the source and the destination state; the admissible class you cite is written with weights depending on the source alone. Are the two compatible, and if the citation is only to norm equivalence, should it not say so? I find the conservation identity fails by up to forty-five per cent under source-only weights.

Q-P06-02 Theorem 5.2 assumes the fine family is flat, and Proposition 6.5 exhibits a charged fine family for which production is nonzero while the coarse deficit vanishes. Should the projection-freeness claim of the abstract and of §5 not carry that hypothesis?

Q-P06-03 What is $h_2(V^\lambda)$ for Example 3.2? I obtain 0.0265 against a deficit norm of 0.0276. Would a charge column in Table 1 not make the paper's headline quantity concrete, and give the programme its first calibration of the constants in (7)?

Q-P06-04 Can L_*, C_* and θ be bounded for $n = 2$ on a three-point grid? Every quantitative charge statement in the paper and its companion rests on constants that are never estimated.

Q-P06-05 In what sense is production "non-negative"? Proposition 6.5 makes it cancel the transmitted deficit exactly.

Q-P06-06 Which dilation category is declared? Equation (3) holds in the time-inhomogeneous category and fails in the time-homogeneous one.

Q-P06-07 In Proposition 6.5 the coarse family is exactly flat. Is the lumped process Markov? I compute that it is not, by exactly $\frac{1}{24}$. What then is "destroyed entirely" by the interference, and should "memory" not be qualified as two-time composition memory throughout?

Q-P06-08 Can the nearest flat family fail to be unique? Take $n = 2$ with uniform laws and the family (I, I, swap): the flat families with one-step kernels $(J/2, I)$ and $(I, J/2)$ are both at squared distance exactly 2, and their residuals differ. In what sense is $V^\lambda - W^*$ canonical?

Q-P06-09 Theorem 5.2(iii) quantifies over block-supported stochastic sections. Does that class contain Mori's linear projection onto a chosen span of observables, or a Wiener projection? If not, in what sense is the dependence on the projection dissolved rather than side-stepped?

Q-P06-10 What is the standard definition of weak lumpability, and does the zero set of production meet it? Your own Corollary 4.6 shows it does not.

Q-P06-11 Does failure of divisibility imply backflow of information in an operational distinguishability sense? The known relation runs the other way: divisibility implies no backflow, and the converse fails.

Q-P06-12 How does the orthogonal splitting of Lemma 2.3(c) relate to the Pythagorean decomposition established for data-processing of Markov chains in information geometry, where a projection onto the lumpable family plays the role your compression plays here? Is your production term the object that has no counterpart there?

Q-P06-13 Open problem (iv) proposes a test of zero production. How does it relate to the Chapman–Kolmogorov test already used to validate reduced Markov descriptions, and does that practice supply the null behaviour you describe as unexplored?

Q-P06-14 Open problem (i) asks for the geometry deciding the sign of $\langle \Phi_\lambda, \mathcal{P}_\lambda \rangle$. Does the dial of Remark 6.6 generalise: is the production always independent of the component of the fine deficit that is block-measurable, so that the cross term is linear in that component and the sign question is one-dimensional?

Q-P06-15 Can the balance law be stated for the full profile h_d rather than the dimension-matched level, as open problem (ii) proposes, given that the model sets at $d > n$ are closures of compressions rather than zero sets of a polynomial system?

Q-P06-16 Does the severed component Ξ_λ have an interpretation as lost information, and is its norm related to any of the established quantities measuring what a lumping destroys?

3 Questions of the portfolio-level evaluation

Source: portfolio-level technical evaluation, version 2, 4 September 2026 (Exhibit H, 02_technical_evaluations), Section 7. Question 7 concerns Paper 6; Question 9 concerns all six papers.

Q-PF-07 **W6 charge-level results.** Give explicit constants, or a worked bound, for the Łojasiewicz comparison (7) in at least one of the four worked families, so that the charge-level balance of Corollary 4.7 is quantitative rather than qualitative.

Note. Q-P06-03 and Q-P06-04 bear on the same constants.

Q-PF-09 **Deposit venue.** State whether the works will be deposited on arXiv (math-ph / math.PR / quant-ph).